Properties of Z transforms

1. Time shifting

$$\{x(n)\} \xrightarrow{Z} X(z) \quad ; \quad \text{ROC } R$$

$$x_1(n) = x(n - N_0)$$

$$\{x_1(n)\} \xrightarrow{Z} X_1(z) = z^{-N_0} X(z) \; ;$$

ROC R except possible addition/deletion of $z=0$ (or) $z=\infty$

Proof:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n-N_0) z^{-n}$$

$$\boxed{l = n - N_0} \quad = \sum_{l=-\infty}^{\infty} x(l) z^{-(l+N_0)}$$

$$= z^{-N_0} \underbrace{\sum_{l=-\infty}^{\infty} x(l) z^{-l}}_{X(z)}$$

$$X_1(z) = z^{-N_0} X(z)$$

2. Linearity

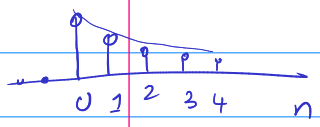
$\{x_1(n)\} \xrightarrow{Z} X_1(z)$ with ROC R_1

$\{x_2(n)\} \xrightarrow{Z} X_2(z)$ with ROC R_2

$\{\alpha x_1(n) + \beta x_2(n)\} \xrightarrow{Z} \alpha X_1(z) + \beta X_2(z)$

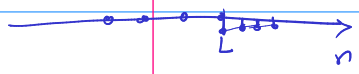
with ROC containing $R_1 \cap R_2$ (atleast as big as)

Example:



$x_1(n) = a^n u(n)$

$$X_1(z) = \frac{1}{1 - az^{-1}} \quad ; \quad |z| > |a|$$



$x_2(n) = -a^n u(n-L)$

$$x_2(n) = -\frac{a^n \times a^{-L}}{a^{-L}} u(n-L)$$

$$= -a^L \cdot (a^{n-L}) u(n-L)$$

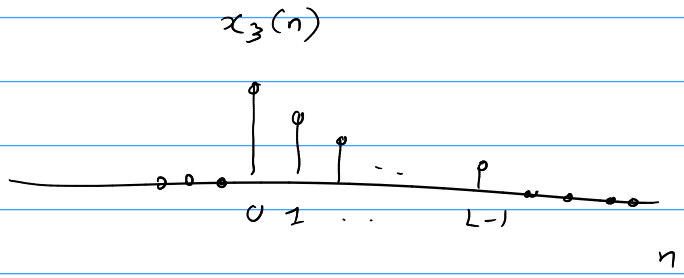
$$= -a^L \cdot x_1(n-L)$$

$$X_2(z) = -a^L \cdot X_1(z) \cdot z^{-L}$$

$$X_2(z) = -z^{-L} a^L \frac{1}{1 - az^{-1}} \quad ; \quad \text{ROC } |z| > |a|$$

$$x_3(n) = x_1(n) + x_2(n)$$

$$= \begin{cases} a^n & ; \quad 0 \le n \le L-1 \\ 0 & ; \quad n \ge L \end{cases}$$



$x_3(n)$ is finite length causal signal

ROC is entire z-plane except $z=0$

Linearity:

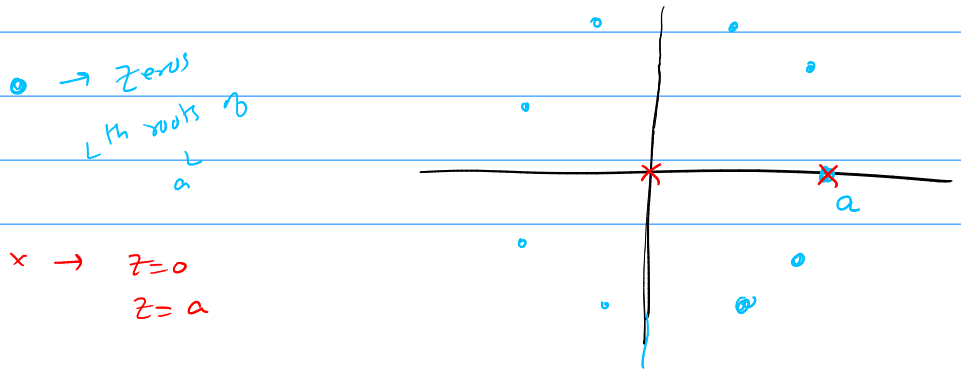
$$X_3(z) = X_1(z) + X_2(z)$$

$$= \frac{1}{1-az^{-1}} - \frac{z^{-L}a^L}{1-az^{-1}}$$

$$X_3(z) = \frac{z^L - a^L}{z^{(L-1)}(z-a)} \quad ; \quad \text{ROC } R: |z|>0$$

Zeros: $z^L - a^L = 0$ $\quad z^L = a^L$

$z \Rightarrow L^{th}$ root of a^L



Pole - zero cancellation occurs at $z = a$

$z = a$ is no longer a pole or zero.

$(L-1)$ zeros

1 pole @ $z = 0$

